

LENANA SCHOOL 2025 PREDICTION



POSSIBLE KCSE EXAM MATHEMATICS

121/2

PAPER 2

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- (a) Write your name, admission number and class in the spaces provided above.
- (b) Sign and write the date of examination in the spaces provided above.
- (c) This paper consists of **two** sections; **Section I** and **Section II**.
- (d) Answer **all** the questions in **Section I** and any **five** questions from **Section II**
- (e) **Show all the steps in your calculations, giving your answers at each stage in the spaces provided below each question**
- (f) Marks may be given for correct working even if the answer is wrong.
- (g) Non-programmable silent electronic calculators and KNEC Mathematical tables may be used, except where stated otherwise.

FOR EXAMINER'S USE ONLY

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

Grand
Total

SECTION 1 (50 MARKS)

For Marking Schemes Contact 0746 222 000 / 0742 999 000

Answer all questions in this section.

1. Solve for x in the equation $2\sin^2 x - 1 = \cos^2 x + \sin x$ for $0^\circ \leq x \leq 360^\circ$ **(3marks)**

2. (a) Expand $\left(1 + \frac{3}{x}\right)^5$ up to the fifth term **(2marks)**

- (b) Hence use your expansion to evaluate the value of $(2.5)^5$ to 3 d.p. **(2 marks)**

3. Complete the table below for $y = 8 - 2x - x^2$ for $-4 \leq x \leq 2$.

x	-4	-3	-2	-1	0	1	2
y							

Hence use trapezium rule with six strips to find the area of the region bounded by the curve and the x- axis.

(3marks)

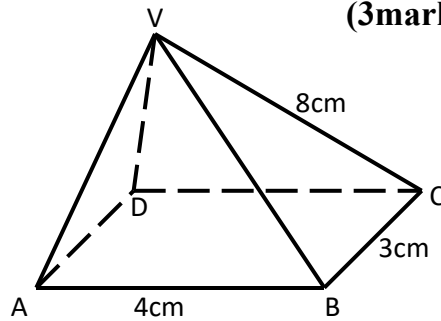
4. Make p the subject of the formula

(3 marks)

$$e = \sqrt{\frac{p-3u}{y-3xp}}$$

5. The figure below shows a rectangular based right pyramid. Find the angle between the planes ABCD and ABV.

(3marks)



6. An object A of area 10cm^2 is mapped onto its image B of area 60cm^2 by a transformation whose matrix is given by $P = \begin{Bmatrix} x & 4 \\ 3 & x+3 \end{Bmatrix}$. Find the possible values of x. (3 marks)

7. Find the value of x in the equation

$$\log_{10} 5 - 2 + \log_{10} (2x + 10) = \log_{10} (x - 4)$$

(3marks)

8. The data below shows marks obtained by 10 students in a test.

71, 55, 69, 45, 65, 57, 71, 82, 55, 50. Calculate the standard deviation using an assumed mean of 60.

(3marks)

9. Evaluate by rationalizing the denominator and leaving your answer in surd form. **(3marks)**

$$\frac{\sqrt{8}}{1 + \cos 45^\circ}$$

10. The position vectors of points A and B are $5i + 4j - 6k$ and $2i - 2j$ respectively. A point X divides AB in the ratio 3: 5. Find the coordinates of X. **(3marks)**

11. A closed box has a square base of side x metres and its height h metres. The total surface area of the box is 24m^2 .

(a) Find the expression of h in terms of x .

(b) Hence find the value of x that would make the volume of the box maximum. **(4 marks)**

12. M varies directly as D and as the cube of V . Calculate the percentage change in M when V is increased by 10% and D is reduced by 10%. **(3marks)**

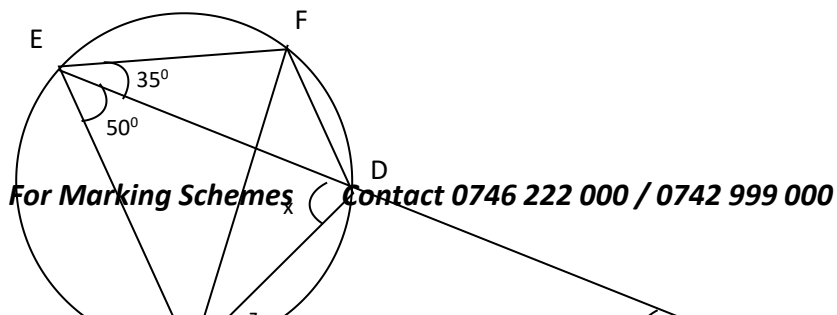
13. Find the value of t if the gradient of the graphs of the functions $y = x^2 - x^3$ and $y = x - tx^2$ are equal at $x = \frac{1}{3}$. (3marks)

14. The image of a point A , under the transformation represented by the matrix

$$T = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} \text{ is } A' (-2, 4) \text{ Find the coordinates of } A \quad (3\text{marks})$$

15. In the figure below, ABC is a tangent at B and CDE is a straight line.

$$\angle BED = 50^\circ, \angle DEF = 35^\circ \text{ and } \angle ECB = 25^\circ$$



Calculate the values of x and z.

(2marks)

16. The equation of a circle is given by $4x^2 + 4y^2 - 8x + 2y - 7 = 0$

Determine the coordinates of the centre of the circle.

(3marks)

SECTION II (50 MARKS)

*Answer **only five** questions from this section*

17. (a) Using a ruler and a pair of compasses only construct triangle ABC in which $AB = 6\text{cm}$, $BC = 5.5\text{cm}$ and angle $ABC = 60^\circ$. Measure AC.

(3marks)

(b) On the same side of AB as C, determine the locus of a point P such that $\angle APB = 60^\circ$

(2marks)

(c) Construct the locus of R such that $AR = 3\text{cm}$

(1mark)

(d) Identify the region T such that $AR \geq 3\text{cm}$ and $\angle APB \geq 60^\circ$ by shading the unwanted part.

(2marks)

(e) Determine point Q such that area of AQB is half the area of ABC and that $\angle AQB = 60^\circ$.

(2marks)

18. A sequence is formed by adding corresponding terms of an AP and GP. The first, second and third terms of the sequence formed are 14, 34 and 78 respectively.

(a) Given that the common ratio of the GP is 3;

(i) Find the first term of the AP and GP and the common difference of the AP. **(2marks)**

(ii) Find the sixth term and the sum of the first six terms of the sequence. **(3marks)**

(b) The second and third terms of a geometric progression are 24 and $12(x + 1)$ respectively. Find the whole number value of x and hence the first term given the sum of the first three terms of the progression is 76. **(5marks)**

19. Income tax rate are as shown below.

Income (k£ p.a)	Rate (Ksh per £)
1- 4200	2
4201 - 8000	3
8001 - 12600	5
12601 – 16800	6

16801 and above	7
-----------------	---

Omari pays Sh. 4000 as P.A.Y.E per month. He has a monthly house allowance of Ksh. 10800 and is entitled to a personal relief of Ksh. 1,100 per month. Determine;

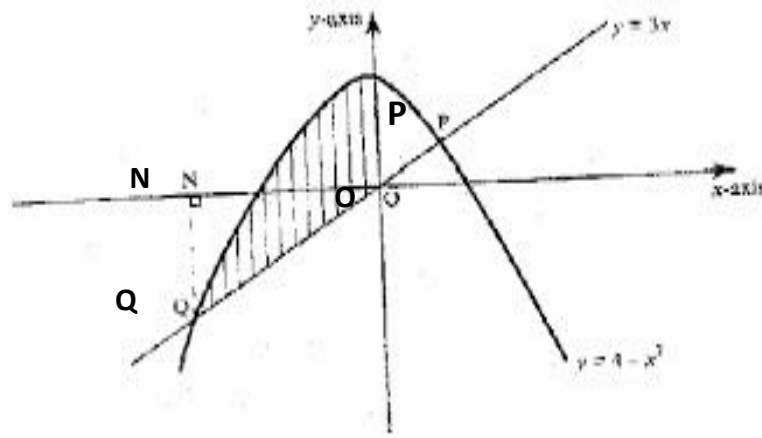
(i) his gross tax p.a in Ksh **(2marks)**

(ii) his taxable income in k£ p.a **(4marks)**

(iii) his basic salary in Ksh. p.m **(2marks)**

(iv) his net salary per month **(2marks)**

20. The diagram below shows a sketch of the line $y = 3x$ and the curve $y = 4 - x^2$ intersecting at point P and Q.



(a) Find the co-ordinates of P and Q

(4marks)

(b) Given that QN is perpendicular to the x-axis at N, calculate

(i) the area bounded by the curve $y = 4 - x^2$, the x-axis and line QN.

(2marks)

(ii) the area of the shaded region that lies below the x - axis

(2marks)

(iii) the area of the region enclosed by the curve $y = 4 - x^2$, the line $y = 3x$ and the y-axis **(2marks)**

21. The positions of two towns A and B are $(50^{\circ}\text{N}, 45^{\circ}\text{W})$ and $(50^{\circ}\text{N}, K^{\circ}\text{W})$ respectively. It takes a plane 5 hours to travel from A to B at an average speed of 800 knots. The same plane takes $1\frac{1}{2}$ hours to travel from B to another town C at the same average speed. Given that C is to the north of B, calculate to the nearest degree,

(a) The value of K

(4marks)

(b) The latitude of C

(3marks)

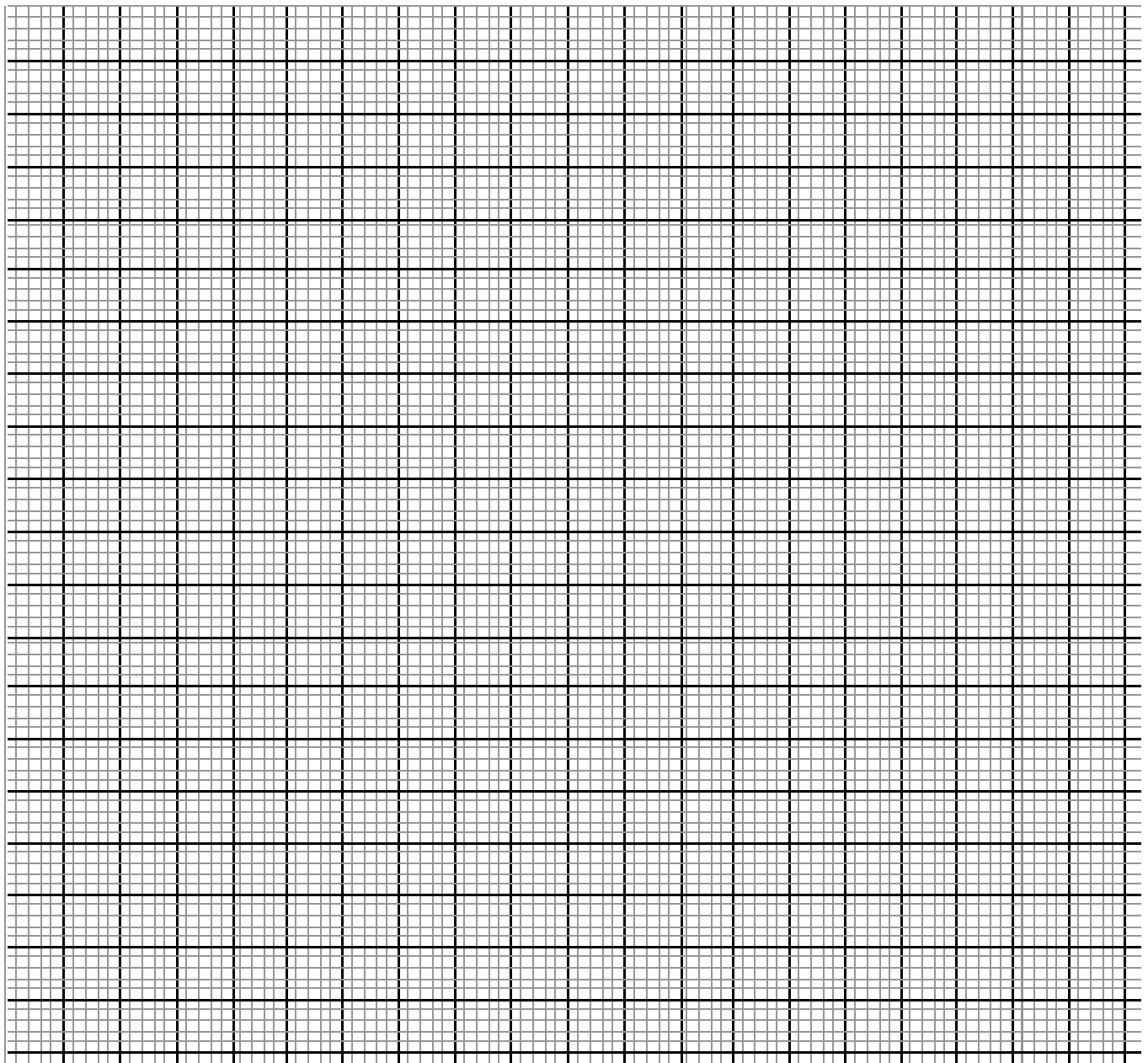
(c) If the plane started from A at 9.00am and flew to C through B, find the local time at C when the plane arrived there.

(3marks)

22. (a) Complete the table below for the equation $y = x^3 + 2x^2$ to 2 d.p **(2marks)**

X	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5
$2x^2$	18	12.5	8	4.5	2		0		1	4.5
X^3	-27		-8		-1		0		1	
y	-9		0		1		0		2	

- (b) On the grid provided, draw the graph of $y = x^3 + 2x^2$ for $-3 \leq x \leq 1.5$. Take a scale of 2cm to represent 1 unit on the x- axis and 1cm to represent 1 unit on the y – axis. **(3marks)**



(c) Use your graph to solve

(i) $x^3 + 2x^2 = 0$

(2marks)

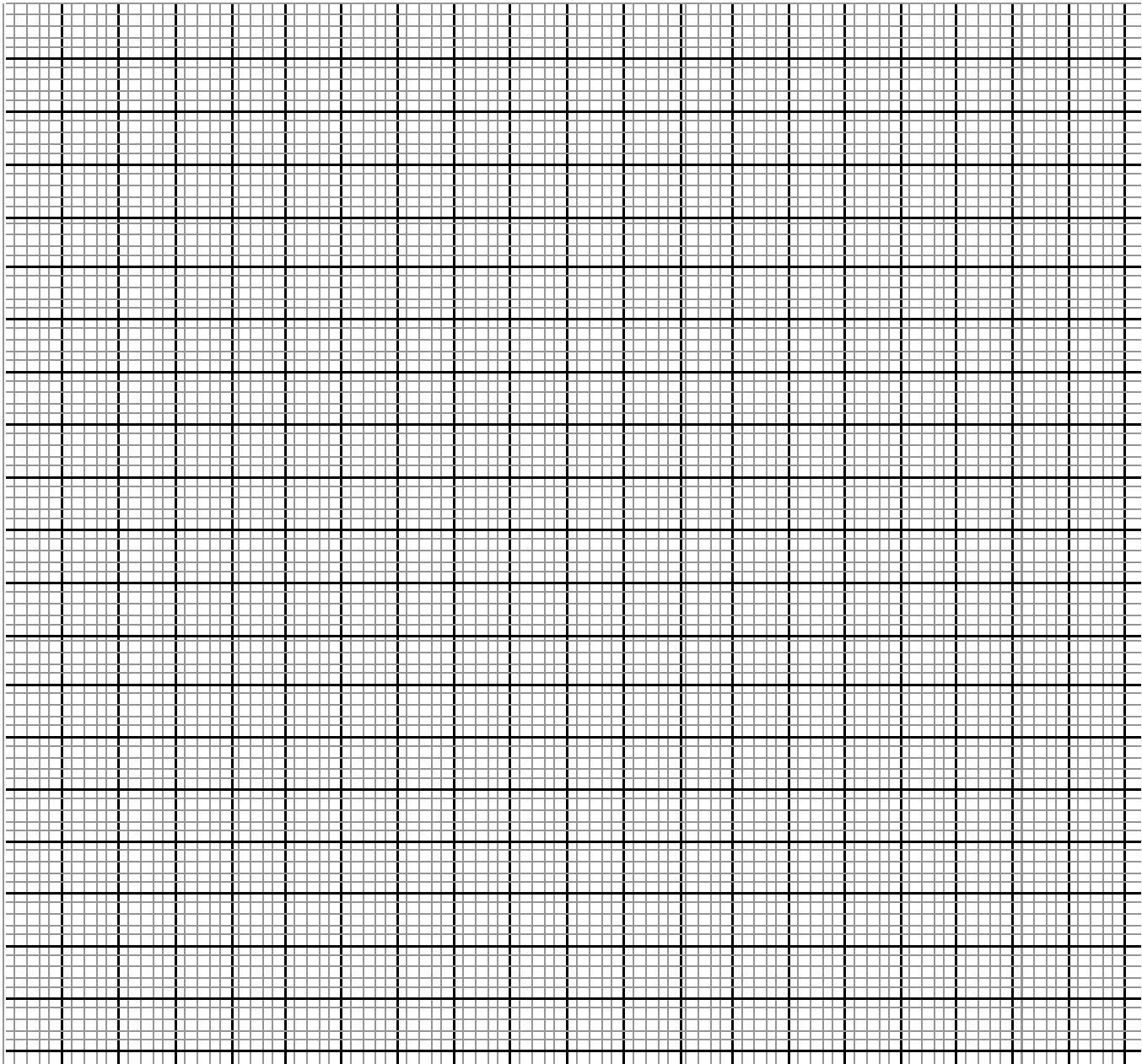
(ii) $x^3 + 2x^2 - x - 2 = 0$

(3marks)

23. The table below shows the marks obtained by 47 students in a mathematics test.

Marks	31 – 35	36 – 40	41 – 45	46 – 50	51 – 55	56 - 60
No. of candidates	4	6	12	15	8	2

(a) On the grid provided, draw a cumulative frequency curve. (3marks)



(b) Use your graph to estimate

(i) The median

(2marks)

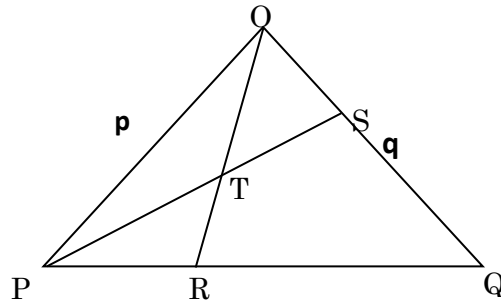
(ii) The semi interquartile range

(2marks)

(c) In order to pass the test a candidate had to score more than 40 marks. Calculate the percentage of candidates who passed.

(3marks)

24. In the triangle OPQ below, $\mathbf{OP} = \mathbf{p}$ and $\mathbf{OQ} = \mathbf{q}$. R is a point on PQ such that $PR: RQ = 1 : 3$ and $5\mathbf{OS} = 2\mathbf{OQ}$. PS intersects OR at T.



(a) Express in term of $\mathbf{p}$ and $\mathbf{q}$

(i) $\mathbf{OS}$

(1mark)

(ii) $\mathbf{PQ}$

(1mark)

(iii) $\mathbf{OR}$

(2mark)

(b) Given that $\mathbf{OT} = h\mathbf{OR}$ and $\mathbf{PT} = k\mathbf{PS}$. Determine the values of h and k . (6marks)